

University of Barishal
Dept. of Computer Science and Engineering
Course Code: CSE-3205:; Course Title: Artificial Intelligence
Admission Session: 2021-22; Exam: Mid Term; Marks: 20; Time: 60 min

1. A delivery robot starts from **S** and wants to reach the destination **G**. The road connections are described as follows: **6**

- S is directly connected to A and B
- A is directly connected to C
- B is directly connected to C
- C is directly connected to G
- All roads have equal cost

The heuristic values are:

Node	h(n)
S	3
A	2
B	1
C	1
G	0

a) First, draw the graph using the above description.
b) Show the node expansion order using
   i) BFS
   ii) Greedy Best-First Search (GBFS)
c) Identify which search is uninformed and which is informed, and analytically explain why GBFS may expand fewer nodes but does not guarantee the shortest path.

2. A university website has the following structure: **8**

- The root page is Home
- Home links to Admission, Academics, and Contact
- Admission links to Form and Scholarship
- Academics links to Departments and Syllabus

A student is searching for the page Syllabus. Now consider the following question:
a) First, draw the search tree from the given description.
b) Write the visiting order of pages using
   i) BFS
   ii) DFS (left-to-right order)
c) Analytically explain:

- which search guarantees the shallowest solution, and
- which search is more memory-efficient.

3. A rescue drone searches for a missing hiker. **6**

The search space is described as follows:

- Each location leads to exactly 4 new locations
- The starting point is the root
- The hiker is guaranteed to be at depth 6

You have to solve the following questions:
a) First, draw a small portion of the search tree (up to depth 2) based on the description.
b) Explain why Iterative Deepening Search (IDS) is still an uninformed search, even though it repeats node expansions.
c) Compare BFS and IDS for this problem in terms of memory usage and practical suitability for a battery-limited rescue drone.

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